# WILLIAM ELIAS

## SOFTWARE & SECURITY AUTOMATION ENGINEER

*Security Automation* | *Credential Remediation* | *OAuth / JWT* | *Python* | *Go* | *C#/.NET*
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## PROFESSIONAL SUMMARY

Software and automation engineer building security tooling and automation in Python, plus developer tools and internal software in Go and C#/.NET, with software and automation work since 2020 and production and infrastructure experience dating to 2015. Builds Python tooling that scans source code and Git history for exposed credentials, plus Git-history remediation tooling with interchangeable rewrite engines. Designed a scoped OAuth 2.1 resource server with JWT/JWKS validation; builds per-module access policies and retained audit history into internal C#/.NET software. M.S. Cyber Defense in progress at Dakota State University, expected Winter 2027.

## PROFESSIONAL EXPERIENCE

**Stellantis Financial Services US** Feb 2023 - Present

*Senior Production Support Engineer* | *DevOps & Automation* | *Auburn Hills, MI · Hybrid*

Promoted September 2026; previously Production Support Engineer

- Built Python tooling to scan source code and Git history for exposed credentials and used BFG Repo-Cleaner to scrub historical records across 100+ repositories. A measured scan processed 2,832 files and identified 67 distinct exposed secrets in approximately 25 seconds; remediation records are complete.
- Designed a scoped OAuth 2.1 resource server with JWT/JWKS validation, content allowlisting, and explicit token-validation-bypass defenses.
- Built and maintain an internal .NET 8 Blazor Web App with Interactive Server, C#, ASP.NET Core, SQL Server, and SQLite for production support, with per-module access policies, database-backed workflows, retained audit history, live configuration/object diffing, pre-change/pre-CAB visibility, and service-desk request tracking.
- Built Python/FastAPI dashboards and operational tooling separate from the Blazor portal, spanning Azure DevOps REST API automation, failed-payload triage, scheduled query-to-email reporting, XML integration-log parsing, ranked engineering search, automated RCA, reusable operations CLI functionality, and recurring production-support workflows; standardized the team on uv for Python dependency and tooling management.
- Designed and delivered a Go-based self-service deployment CLI covering the build-to-rollback lifecycle with guided setup, Azure DevOps REST integration, config parsing, pipeline generation, build/push/deploy, rollback, status, audit, and post-deployment validation.
- Directed a CI/CD standardization program across a 60-repository, 104-application internal estate, building reusable pipeline/template automation and 56 Azure DevOps build/release definitions across 28 standardized application repositories.
- Executed and validated builds across all 28 repositories (27 passed, with the remaining failure diagnosed and logged); dry-run validated representative deployment paths for 25 of 28, with the remaining checks blocked by upstream/environment or tooling conditions. Reviewed and classified 109 existing definitions, identifying 95 legacy and 14 already-aligned definitions, and resolved several latent delivery defects spanning path-handling assumptions, configuration drift, and false-success states.
- Maintain the release-tracking process for a recurring, roughly weekly production release cadence and participate directly in production deployments. Authored and maintain a reusable 22-query KQL library used across internal monitoring and troubleshooting automation.

**HBK - Hottinger Brüel & Kjær** Jul 2020 - Feb 2023

*DevOps Engineer - Python Automation* | *Southfield, MI · Remote*

- Created Python/Tkinter XML-output automation that reduced data-processing time by 60%; automated SQL output processing to save more than 40 hours per month.
- Developed database-configuration tools to streamline customer onboarding and reduce repetitive environment setup.

## CORE EXPERTISE

**Security & Identity:** OAuth / OIDC, Credential Remediation, Git History Remediation, Audit Controls
**Automation & Developer Tooling** *(Python/PowerShell professionally; Go for deployment tooling; uv for Python standardization)***:** Python, PowerShell, REST APIs, Go, uv, Developer Tooling
**Software & Backend:** Python, FastAPI, C#, .NET, Blazor, ASP.NET Core, REST APIs, SQL Server, SQLite

**DevOps & Delivery** *(Azure DevOps professionally; containerization proof of concept)***:** Azure DevOps, CI/CD, Release Automation, YAML Pipelines, Git, Containerization

## SELECTED ENGINEERING HIGHLIGHTS

### Git-History Credential Remediation (Python)

- Git-history remediation tooling combines repository discovery, BFG Repo-Cleaner, reflog expiry, and coordinated cleanup of exposed credentials, supporting single-repository and whole-project modes with interchangeable rewrite engines.

*Tech: Python, Git, BFG Repo-Cleaner, Credential Remediation, Git History Remediation*

### Scoped OAuth 2.1 Resource Server (Python)

- Design for a read-only resource server integrating an enterprise AI assistant with internal engineering documentation: validates issuer and audience claims through JWT/JWKS, restricts signing algorithms, tests rejection of unsigned / alg=none tokens, and enforces scoped content allowlisting.

*Tech: Python, OAuth 2.1, JWT, JWKS, REST APIs*

### Internal Web Application with Access Policies & Audit (C#/.NET)

- Change-safety features in the modular .NET 8 Blazor Web App include DBA approve-only deployment scripts, live configuration/object diffing for pre-CAB and pre-deployment review, per-module access policies, and retained audit history; integrated service-desk request tracking keeps operational workflows traceable.

*Tech: C#, .NET 8, Blazor, ASP.NET Core, SQL Server, SQLite*

## SELECTED OPEN-SOURCE ENGINEERING

### Password Arena

github.com/howlcipher/password_arena

- Built a local attacker/defender simulation for synthetic password-strength experiments with bounded budgets, reproducible test modes, metrics, and audit reports.
- Added synthetic-only safety boundaries, CSPRNG generation, and pytest, mypy, and Ruff quality checks.

### HowlPlane

github.com/howlcipher/howlplane

- Built a Python/Go engineering control plane that routes work across coding agents, reconciles independent reviews, and runs deterministic verification.
- Added durable evidence records, human approval gates for consequential actions, and CI evidence bound to the pull-request head commit.

## EARLIER EXPERIENCE

**QA Developer, Intrepid Control Systems** Mar 2020 - Apr 2020

- Wrote Mocha tests and debugged APIs across software and hardware integrations.

**Network Engineer, Project Worldwide** Jan 2020 - Mar 2020

- Configured Cisco/Meraki infrastructure and executed multi-state network migrations with minimal downtime.

**Network Standards Engineer, Ford Motor Company** Sep 2019 - Dec 2019

- Implemented global network standards, documentation, and IP-management systems.

**IT Network Engineer, Trendset Communications Group** Sep 2015 - Jun 2019

- Maintained network standards, configurations, and documentation; supported critical production infrastructure.

## EDUCATION & CERTIFICATIONS

M.S. Cyber Defense (In Progress, Second Year) - Dakota State University (Expected Winter 2027)
B.S. Information Technology - Colorado State University Global Campus
B.B.A. Business Administration - Rochester College
CCNA (Previously Held) - Cisco Networking Academy (2014 - 2017)